Part A - 2 Mark

1. **Discuss the significance of back emf.**
2. **Name any four applications of DC shunt motors.**
3. **What is the function of a commutator in a DC generator?**
4. **A 25 kVA, single-phase transformer has 250 turns on the primary and 40 turns on the secondary winding. The primary is connected to 1500-volt, 50 Hz mains. Calculate the Primary and Secondary currents on full load.**
5. **State any two applications of capacitor start induction motor.**
6. **Draw the graph that depicts the correlation between the speed and armature current of a shunt motor.**
7. **Mention the types of breaking of DC motors.**
8. **What is the transformation ratio?**

Part B – 13 Marks

1. **With the necessary equations, illustrate the types of DC motors.**
2. **With the necessary equations, illustrate the types of DC generators.**
3. **Differentiate between core type and shell type transformer**?
4. **With a neat sketch, Explain the construction of DC machines.**
5. **Explain the working principle of the transformer and derive the EMF equation of the Transformer.**
6. **Describe with neat sketches the constructional details of a salient pole-type alternator.**
7. **With neat sketches, Explain the types of transformers based on construction and voltage.**
8. **A 6600/440V Single phase 600 KVA transformer has 1200 primary turns. Find**
   (i) Transformation ratio
   (ii) Secondary turns
   (iii) Voltage/turn
   (iv) Secondary current when it supplies a load of 400 kW at 0.8 p.f. lagging.

Part C – 15 Mark

1. I. **Derive the EMF equation of the DC generator.** (9)

   II. **Calculate the voltage induced in the armature winding of a 4-pole, lap wound DC machine having 728 conductors and running at 1800 rpm. The flux per pole is 30 mWb. (6).**

2. I. **Explain the various methods of excitation of DC machines. (10)**
   II.**Draw the OCC characteristics and external characteristics of the DC generator. (5)**